

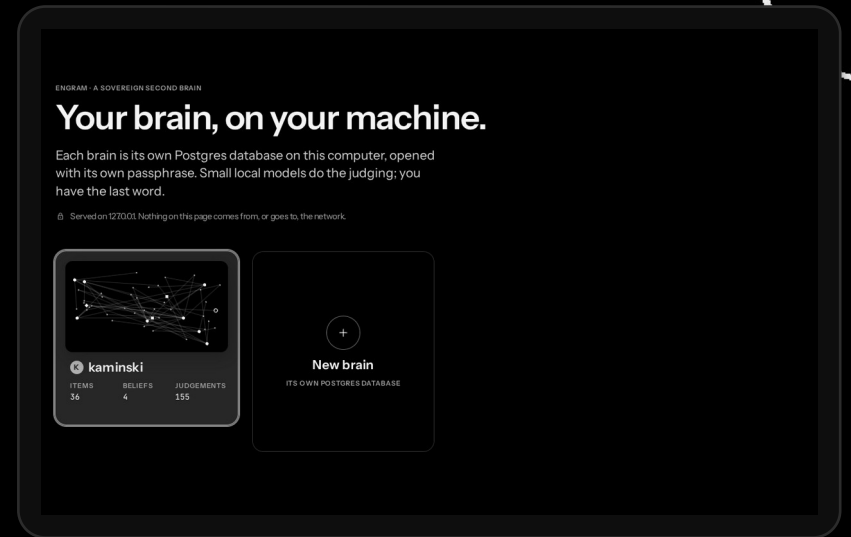
Omnitrix

A local AI secretary that remembers, plans and acts, only with your OK.

Brain: engram

Track 1 Sovereign AI

MSRIT Hackathon



The Problem:

Data Overload & Unreliable AI Memory

Context fragmentation



Emails



Invites



Notes

Decisions scattered across emails,
calendar invites and notes

Broken AI memory

97.8%

junk in a Mem0 audit

41%

Graphiti facts
invalidated

3 of 4 audited invalidations were wrong

Privacy leaks

7.5%

of queries leak private data in local +
cloud delegation (PAPILLON)

Our answer: a personal AI.



Personal AI

learns how you judge



Locally hosted

zero cloud calls



Fine-tuned LLM

a 1.7B trained on you



Society of LLMs

fast small, slow large, you



Second brain

memory with sources

How it decides

Escalate only
when unsure ↗

S0 • Code

10 of 55 are code

S1 • Small judge

42 of 55 start here

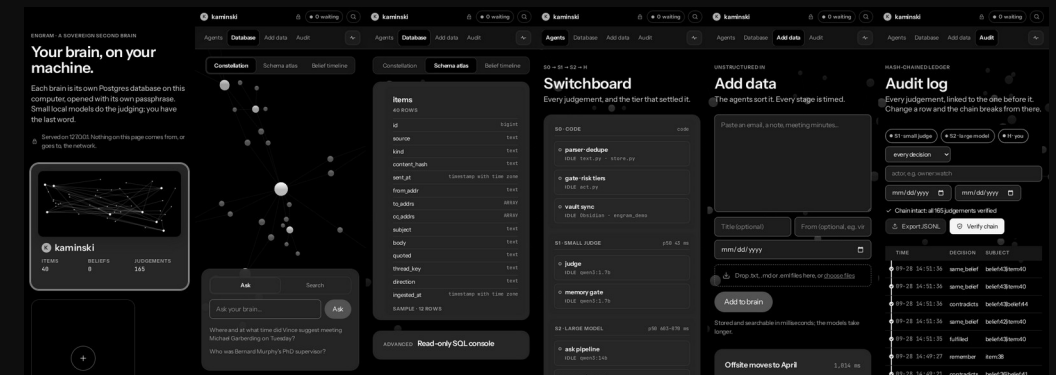
S2 • Large model

1 of 55 starts here

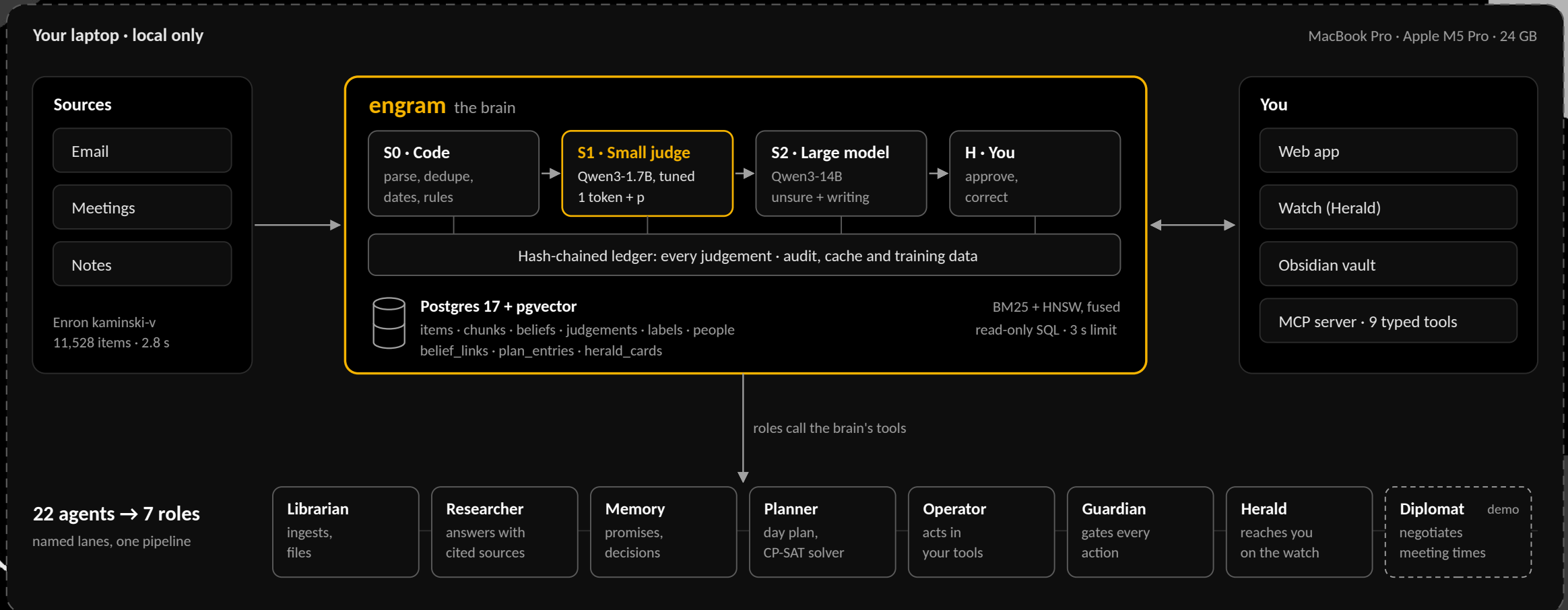
H • You

the last word

The engram web app, on a phone

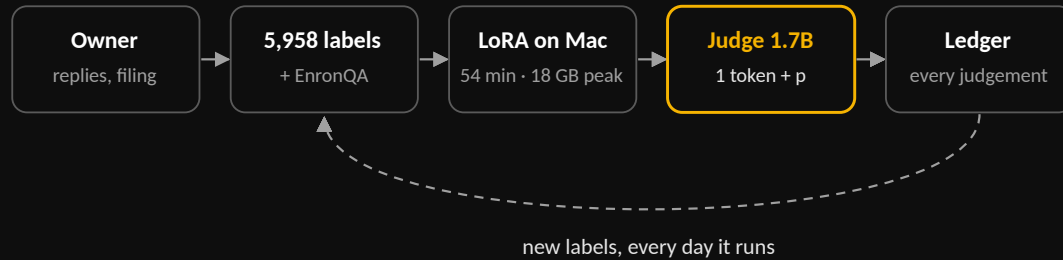


The brain and its society of agents.

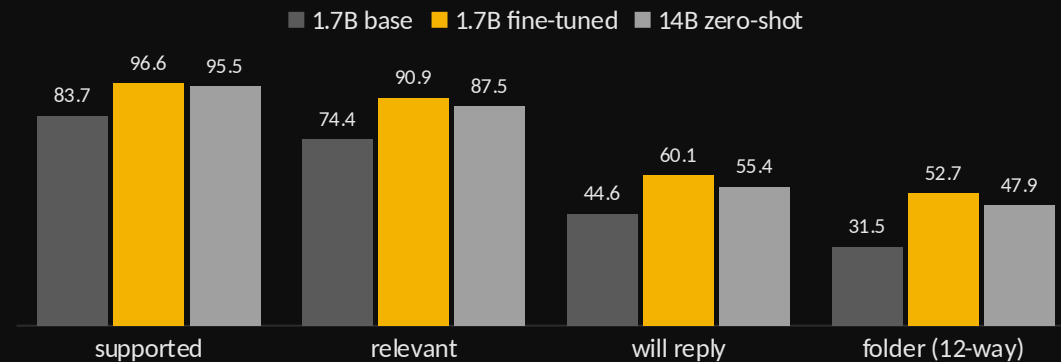


Fine-tuned on you, approved on your wrist.

Skills are trained, never facts · no hand labels · no cloud teacher



Held-out accuracy, %



Gate pauses
Guardian holds the action

Long-poll
watch waits on the Mac

Watch buzzes
an approval card

You tap
Approve or Deny

Ledger
decision recorded

gate → ledger **19 s**

HERALD

Approve?

email.reply_email

Deny

Approve

mDNS + HMAC

Galaxy Watch 5 · Wear OS 5 · Kotlin
direct Wi-Fi to the Mac · no cloud push

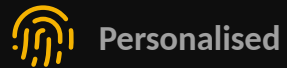
Impact: private, personal, faster.



Private

0

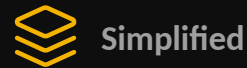
cloud calls; every model runs locally



Personalised

96.6%

fine-tuned 1.7B vs 95.5% for the 14B



Simplified

7 roles

and one watch tap to approve



Time

807 s → 255 s

memory gate on 60 emails



Cost

3–5×

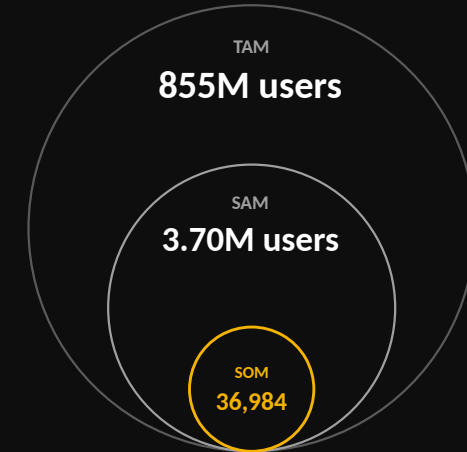
faster than the 14B · 934 MB · no API costs



Accuracy

40.0% → 6.7%

harmful side effects on 60 sandbox tasks



TAM 1.14B knowledge workers × 75% using AI = 855M × ₹23,988/yr ≈ ₹20.5 lakh crore

SAM India's advocates, CAs, doctors and startups: 40.2 lakh × 92% using AI = 3.70M × ₹23,988 ≈ ₹8,872 crore

SOM 1% of SAM in 3 years (our assumption) = 36,984 × ₹23,988 ≈ ₹88.7 crore

Price benchmark: ChatGPT Plus in India, ₹1,999/month. Sources on slide 8.

Feasible today, on one laptop.

Compute

24 GB

one MacBook Pro, Apple M5 Pro

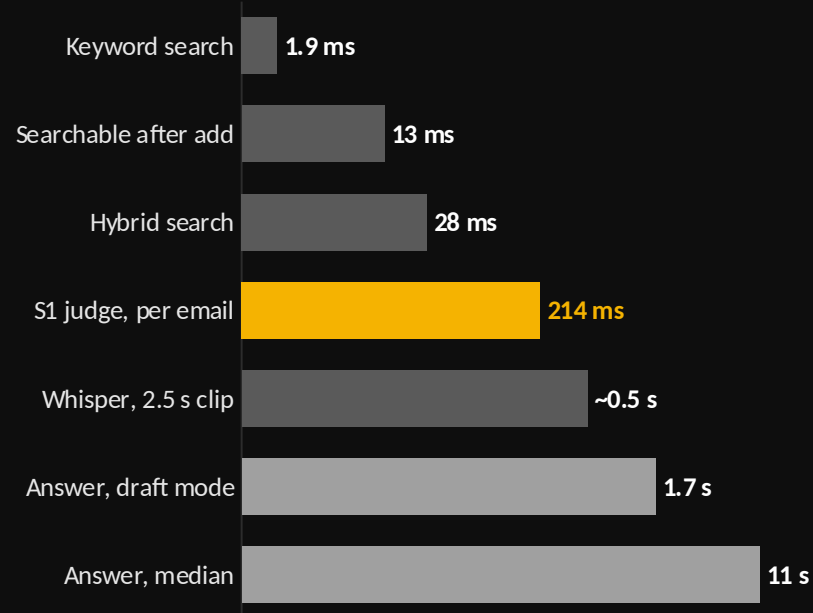
934 MB

fused 4-bit judge

54 min

LoRA training · 18 GB peak

Brain latency log scale



Privacy

 Zero cloud calls

 Hash-chained ledger

 DPDP-ready

up to ₹250 crore

DPDP Rules 2025 penalty, from ~May 2027

Runs today on hardware the owner already has.

References and sources.

Research and evidence

Kim et al. Multi-agent scaling across 180 configurations. arXiv:2512.08296

Cemri et al. MAST: why multi-agent systems fail. arXiv:2503.13657

Zheng et al. Persona prompts don't improve accuracy. arXiv:2311.10054

NATURAL PLAN. arXiv:2406.04520

A solver-checked LLM planner (10% → 93.9%). arXiv:2404.11891

NVIDIA. Small Language Models are the Future of Agentic AI. arXiv:2506.02153

PAPILLON (NAACL 2025): local + cloud delegation leaks 7.5%

Mem0 memory audit (mem0 issue #4573); Graphiti invalidation report (graphiti issue #1728)

India's Digital Personal Data Protection Rules, 2025

Data, models and market

Datasets: EnronQA (CC BY 4.0), WorkBench (MIT), QMSum / AMI

Models and tools: Qwen3-1.7B, Qwen3-14B, Whisper large-v3-turbo, MLX, Ollama, PostgreSQL + pgvector, OR-Tools CP-SAT

Market (TAM / SAM / SOM)

Gartner press release, 24 Sep 2019: 1.14B knowledge workers by 2023

Microsoft & LinkedIn, 2024 Work Trend Index: 75% of knowledge workers use AI

Microsoft India, 16 May 2024: 92% of Indian knowledge workers use AI

Ministry of Law & Justice, Rajya Sabha reply (LawChakra, 25 Oct 2025): 20,13,081 advocates

ICAI 76th Annual Report 2024-25: 4,23,105 members

MoS Health, Rajya Sabha reply (1 Apr 2025): 13,86,150 allopathic doctors

DPIIT via PIB, 2 Dec 2025: 1,97,692 recognised startups

TechCrunch, 18 Aug 2025: ChatGPT Plus ₹1,999/month in India



Thank you

Your brain, on your machine.

[**github.com/tharanirmal/omnitrix**](https://github.com/tharanirmal/omnitrix)

Repo



Team